The evidence for such a hole in our own Milky Way galaxy (as of 1993) is suggestive, but far from firm. One key bit of evidence comes from the orbital motions of gas clouds near the center of the galaxy. Infrared observations of those clouds, by Charles Townes and colleagues at the University of California at Berkeley, show that they are orbiting around an object which weighs about 3 million times as much as the Sun, and radio observations reveal a very peculiar, though not strong, radio source at the position of the central object—a radio source amazingly small, no larger than our solar system. These are the types of observations one might expect from a quiescent, 3-million-solar-mass black hole with only a tenuous accretion disk; but they are also readily explained in other ways.

The possibility that gigantic black holes might exist and inhabit the cores of galaxies came as a tremendous surprise to astronomers. In retrospect, however, it is easy to understand how such holes might form in a galactic core.

In any galaxy, whenever two stars pass near each other, their gravitational forces swing them around each other and then fling them off in directions different from their original paths. (This same kind of swing and fling changes the orbits of NASA's spacecraft when they encounter planets such as Jupiter.) In the swing and fling, one of the stars typically gets flung inward, toward the galaxy's center, while the other gets flung outward, away from the center. The cumulative effect of many such swings and flings is to drive some of the galaxy's stars deep down into the galaxy's core. Similarly, it turns out, the cumulative effect of friction in the galaxy's interstellar gas is to drive much of the gas down into the galaxy's core.

As more and more gas and stars accumulate in the core, the gravity of the agglomerate they form should become stronger and stronger. Ultimately, the agglomerate's gravity may become so strong as to overwhelm its internal pressure, and the agglomerate may implode to form a gigantic hole. Alternatively, massive stars in the agglomerate may implode to form small holes, and those small holes may collide with each other and with stars and gas to form ever larger and larger holes, until a single gigantic hole dominates the core. Estimates of the time required for such implosions, collisions, and coalescences make it seem plausible (though not compelling) that most galaxies will have grown gigantic black holes in their cores long before now.